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PROBLEM STATEMENT TITLE: ATMs are widely used for banking, but many people in rural and semi-urban areas face difficulties due to language barriers and low literacy levels, Existing ATMs support limited languages and rely on text-based instructions, making them hard to use for many users. This leads to dependency and lack of access to essential financial services. Therefore, a multilingual, voice-enabled ATM system is needed to provide simple, user-friendly, and inclusive banking access.

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ABSTRACT:

The Multilingual ATM System with Voice Guidance is a web-based application designed to enhance accessibility and usability of Automated Teller Machines for users from diverse linguistic backgrounds. The system supports 15 different languages, allowing users to interact with the ATM interface in their preferred language, thereby reducing language barriers and improving user experience.

The application provides essential banking functionalities such as cash withdrawal, deposit, balance inquiry, fast cash, and mini statement, all within a user-friendly interface that simulates a real ATM environment. A key feature of this system is the integration of voice guidance, which delivers step-by-step instructions in the selected language using speech synthesis technology. This feature is especially beneficial for visually impaired users and first-time ATM users.

The system also includes secure PIN entry through a virtual keypad, structured menu navigation, and receipt generation after transactions. Overall, this project demonstrates how multilingual support and voice assistance can significantly improve the efficiency, accessibility, and inclusiveness of ATM systems.

INTRODUCTION:

Automated Teller Machines (ATMs) have become an essential part of modern banking, providing users with convenient access to financial services such as cash withdrawal, deposit, and balance inquiry. However, many existing ATM systems are limited in terms of language support and accessibility, making it difficult for users who are not familiar with the default language or who require additional assistance.

The Multilingual ATM System with Voice Guidance is developed to address these challenges by offering an inclusive and user-friendly interface. This system supports multiple regional languages, enabling users to perform transactions in their preferred language. In addition, it incorporates voice guidance, which provides step-by-step instructions, making the system easier to use for beginners and visually impaired individuals.

The project simulates a real ATM environment with features such as secure PIN entry, menu-driven transactions, fast cash options, and receipt generation. By integrating multilingual support and audio assistance, this system aims to improve user experience, accessibility, and efficiency in ATM operations.

LITERATURE REVIEW:

The development of Automated Teller Machines (ATMs) has significantly improved banking accessibility and convenience over the years. Traditional ATM systems mainly focus on basic functionalities such as cash withdrawal, deposit, and balance inquiry, but they often lack user-friendly features for people from diverse linguistic and educational backgrounds. Several research works have highlighted the need for more inclusive ATM systems.

Previous studies have focused on multilingual user interfaces, emphasizing the importance of providing regional language support to reduce communication barriers. These systems allow users to interact with machines in their native language, improving usability and reducing errors during transactions. However, many of these implementations are limited to only a few languages and do not fully address accessibility challenges.

Research in the field of human-computer interaction (HCI) suggests that voice-based systems can significantly enhance user experience, especially for visually impaired users and first-time users. Voice guidance systems using speech synthesis technology have been successfully applied in various applications, including navigation systems and virtual assistants, but their integration into ATM systems is still limited.

Some existing ATM models include audio assistance through headphones, but they often require additional hardware and are not widely implemented. Moreover, these systems may not support multiple languages effectively. There is a gap in developing a cost-effective solution that combines both multilingual support and voice guidance in a simple and efficient manner.

The proposed Multilingual ATM System with Voice Guidance addresses these limitations by integrating a wide range of languages along with built-in voice assistance using web technologies. This approach enhances accessibility, improves user interaction, and provides a more inclusive ATM experience compared to traditional systems.

PROBLEM STATEMENT-Design methodology:

PROBLEM STATEMENT:

Automated Teller Machines (ATMs) are widely used for performing essential banking transactions such as cash withdrawal, deposit, and balance inquiry. However, most existing ATM systems are designed with limited language options, typically supporting only one or two major languages. This creates a significant barrier for users who are not familiar with these languages, especially in a linguistically diverse country like India, where people speak different regional languages.

In addition to language limitations, traditional ATM systems often lack proper guidance for users during transactions. First-time users, elderly individuals, and people with low technical knowledge may find it difficult to understand the steps involved, leading to confusion, errors, or incomplete transactions. Furthermore, visually impaired users face even greater challenges due to the absence of effective audio assistance or voice-based interaction in many ATM machines.

Although some advanced ATM systems provide audio guidance through external devices such as headphones, these solutions are not widely available and can increase system complexity and cost. There is also a lack of integration between multilingual interfaces and voice assistance in a single platform.

Therefore, there is a strong need to develop an improved ATM system that combines multilingual support with built-in voice guidance, enabling users to perform transactions easily in their preferred language. The system should provide clear step-by-step instructions, reduce dependency on external help, and enhance accessibility, usability, and inclusiveness for all categories of users.

Design Methodology

The design of the Multilingual ATM System with Voice Guidance follows a user-centered approach to ensure simplicity, accessibility, and efficiency. The system is developed using web technologies such as HTML, CSS, and JavaScript, which provide an interactive and responsive interface similar to a real ATM environment.

1. System Design

The system is divided into multiple modules, including language selection, transaction menu, PIN verification, amount entry, transaction processing, and receipt generation. Each module is designed as a separate screen to ensure smooth navigation and clarity for the user.

2. User Interface Design

A grid-based layout is used to replicate the real ATM display. The language selection screen is designed in a 4×4 grid format, while the transaction menu follows a 3×2 structure for easy access. Large buttons and clear text are used to enhance readability and touch interaction.

3. Multilingual Implementation

The system supports 15 different languages by storing all text content in a structured JavaScript object. Based on the user's selection, the interface dynamically updates all labels, instructions, and messages in the chosen language.

4. Voice Guidance System

Voice assistance is implemented using the Web Speech API (Speech Synthesis). The system provides step-by-step audio instructions in the selected language at every stage, such as language selection, PIN entry, transaction choice, and completion.

5. Input Handling

A virtual keypad is used for secure PIN entry and amount input, simulating a real ATM keypad. This ensures better control and prevents manual keyboard input, improving security and usability.

6. Transaction Flow

The system follows a structured flow:
Language Selection → Transaction Selection → PIN Verification → Operation Execution → Result Display → Receipt Generation

7. Output and Feedback

The system provides both visual and audio feedback for every action. After completing a transaction, a digital receipt is generated displaying transaction details such as date, type, and status.

Development of SYSTEM / prototype:

The Multilingual ATM System prototype is developed using web-based technologies such as HTML, CSS, and JavaScript to simulate the working of a real ATM machine with enhanced accessibility features. The system interface is designed to resemble an actual ATM display, providing a full-screen, touch-friendly experience with clearly structured sections for language selection, transaction menu, PIN entry, amount input, and result display. A 4×4 grid layout is used for displaying 15 different language options, enabling users to easily select their preferred language. Once selected, all interface elements such as instructions, buttons, and messages dynamically change to the chosen language, ensuring ease of understanding for users from diverse linguistic backgrounds.

The system incorporates a voice guidance feature using the Web Speech API, which provides audio instructions at every step of the transaction process. This includes welcome messages, menu instructions, PIN entry prompts, and transaction confirmations in the selected language. This feature significantly improves usability, especially for visually impaired users and individuals unfamiliar with ATM operations. The transaction flow is carefully structured to replicate real ATM functionality, beginning with language selection, followed by transaction selection (Withdraw, Deposit, Balance Inquiry, Fast Cash, and Mini Statement), and then PIN verification. A virtual numeric keypad is implemented for secure PIN entry, ensuring that users interact with the system in a way similar to a physical ATM.

For transactions involving money input, such as withdrawal and deposit, an amount entry screen with a numeric keypad is provided, allowing users to input values easily. The Fast Cash feature includes predefined amounts for quick transactions, along with an “Other” option for custom input. In the case of balance inquiry, the system directly displays the result without requiring an amount entry, while the mini statement option shows recent transaction details without displaying unnecessary fields. After completing a transaction, the

system generates a result message indicating success and provides an option to print a receipt.

The receipt generation module creates a digital receipt formatted similarly to a real ATM slip, including details such as date, transaction type, amount (if applicable), and transaction status. The overall system is tested to ensure smooth navigation, correct language switching, proper voice output, and accurate simulation of ATM operations. This prototype demonstrates a practical and user-centric solution that enhances accessibility, usability, and inclusivity, and it can be further developed into a real-world ATM system with backend integration and security features.

Design validation/ Prototype testing:

The design validation and prototype testing of the Multilingual ATM System are carried out to ensure that the system functions accurately, efficiently, and meets user requirements. The prototype is tested across multiple scenarios to verify the correctness of each module, including language selection, voice guidance, transaction flow, and user interface responsiveness. During validation, all 15 language options are checked to confirm that the interface text updates correctly and that the voice output corresponds to the selected language without errors. The voice guidance system is also tested to ensure clear audio output at every stage, including welcome messages, menu instructions, PIN entry prompts, and transaction confirmations.

Functional testing is performed to validate the complete transaction process, including Withdraw, Deposit, Balance Inquiry, Fast Cash, and Mini Statement operations. Special attention is given to ensure that the balance inquiry does not require amount input and that the mini statement displays only relevant transaction details. The PIN verification module is tested with both correct and incorrect inputs to confirm proper error handling and system response. The keypad functionality for PIN and amount entry is verified to ensure accurate input capture and user-friendly interaction.

User interface testing is conducted to evaluate the layout design, including the 4×4 language grid and 3×2 transaction menu, ensuring that the system is intuitive and easy to navigate. Responsiveness of the interface on different screen sizes, especially laptop displays, is also validated. The receipt generation

feature is tested to confirm that it displays correct transaction details in a proper format similar to real ATM receipts.

Finally, usability testing is performed by simulating real user interactions to assess ease of use, clarity of instructions, and overall user experience. The results of testing indicate that the prototype successfully meets the design objectives by providing a reliable, accessible, and user-friendly ATM interface with multilingual and voice-enabled support.

RESULTS AND DISCUSSIONS:

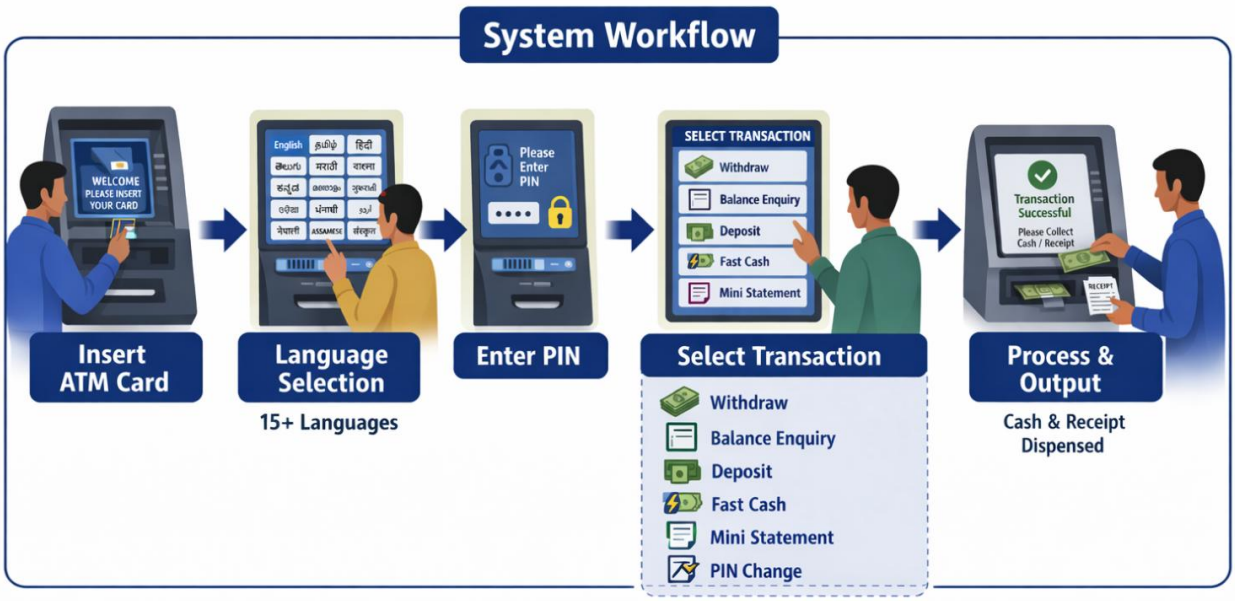
The results of the Multilingual ATM System demonstrate that the prototype successfully achieves its objective of providing an accessible and user-friendly ATM interface. The system effectively supports 15 different languages, allowing users to interact in their preferred language without confusion. During testing, all language selections correctly updated the interface text and provided appropriate voice guidance, which enhanced usability, especially for users who are not comfortable with a single default language. The voice assistance feature performed reliably at each stage of the transaction, guiding users through the process and reducing dependency on prior ATM knowledge.

The transaction modules, including Withdraw, Deposit, Balance Inquiry, Fast Cash, and Mini Statement, were executed smoothly with proper navigation between screens. The PIN verification system ensured basic security simulation, and the keypad interface allowed accurate input for both PIN and amount entry. The Fast Cash feature improved efficiency by offering predefined transaction amounts, while the balance inquiry and mini statement functions displayed only relevant information, making the system more realistic and user-friendly. The receipt generation module produced properly formatted transaction details similar to an actual ATM receipt, enhancing the authenticity of the prototype.

From a usability perspective, the grid-based layout (4×4 for languages and 3×2 for menu options) provided a clear and organized interface, making it easy for users to navigate. The touch-friendly design and large buttons further improved interaction, especially for first-time users. Overall, the system performed consistently across different test scenarios, indicating reliability and effectiveness. The discussion highlights that integrating multilingual support

and voice guidance significantly improves accessibility and inclusivity in ATM systems, making them more adaptable for diverse user groups.

CONCLUSION:

The Multilingual ATM System successfully demonstrates an innovative approach to improving accessibility and usability in ATM interfaces by integrating support for 15 languages along with voice guidance at every step. The system provides a user-friendly, touch-based interface that closely simulates real ATM operations, including transaction selection, PIN authentication, cash handling options, and receipt generation. By allowing users to interact in their preferred language and offering clear audio instructions, the prototype effectively reduces language barriers and enhances ease of use for diverse users, including those with limited literacy or prior ATM experience.

Overall, the project highlights the importance of inclusive design in banking technology and shows how multilingual and voice-enabled systems can significantly improve user experience. The prototype performs efficiently across all tested functionalities and serves as a strong foundation for future development. With further enhancements such as backend integration, real-time banking connectivity, and advanced security features, this system can be implemented in real-world ATM machines to provide a more accessible and user-centric banking solution.